

INFORMATICS INSTITUTE OF TECHNOLOGY

In Collaboration with

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NUTRI SCANNER

(Personalized Nutrition Tracking and Health Risk Prediction System)

Group 10 Project Proposal by

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Declaration

We hereby certify that this project proposal and all the artifacts associated with it is our own work and it has not been submitted before nor is currently being submitted for any degree program.

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I have read the project proposal, and it is in accordance with the approved university project proposal outline.

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1. Introduction

The Nutri Scanner project is an AI-powered nutrition tracking and health risk prediction app designed to help Sri Lankan users easily understand food labels, analyze traditional homemade meals, and manage their diet utilizing computer vision, OCR and NLP, it reads multilingual food labels and analyze meals images, offering personalized nutrition advice based on individual health profiles. The system predicts risk for conditions like diabetes, obesity, and hypertension while tracking user progress over time, making healthy eating more accessible and relevant to the local context. This tailored approach addresses the limitations of global apps that lack support for Sri Lanka's unique dietary patterns and languages, empowering users with accurate AI-driven nutritional insights.

1.1. Problem Domain

Sri Lankan cuisine is diverse and rich, combining large amounts of prepared foods with popular traditional home cooking. Nutrition labeling plays a crucial role in helping people make improved food decisions and averting the epidemic rise of diet-related chronic conditions like diabetes, hypertension, and heart disease. Most Sri Lankan consumers, however, cannot read food labels since they are typically replicated in duplicate at twice the size in small letters and jargon that one cannot decipher. Although nutrition labels are good tools, they are poorly utilized by most, and this limits their potential to healthier consumption (Maduwearachchi, Jemziya, & Rifath, 2024).

To add to this, the majority of the nutrition monitoring apps that are available in the marketplace are developed mostly for Western markets. They're formatted based on English labels and common Western foods, so they don't accommodate Sri Lanka's diverse multilingual food culture as well as traditional recipes. These applications are not suitable for local languages, types of food, or cultures of eating and therefore have not picked up much here. Sri Lanka boasts a mounting public health issue with diet-related illness on the rise. Many children suffer from stunted growth and are not being given proper nutrients, while more adults are gaining weight or developing illnesses like diabetes. These are issues that highlight the necessity for thorough research and more attention. The problem needs to be tackled to create awareness about healthy eating, deliver individualized nutrition counseling, and ultimately empower citizens throughout the country to live healthier lives.

1.2. Problem Definition

In Sri Lanka, most of the population cannot make informed food choices due to deceptive and unreadable food packaging. Processed food tends to have details both in Sinhala as well as English, but the script is very small and filled with technical terms that most consumers are unable to read (Perera, Jayawardena & Arunatilake, 2022 Wickramasinghe, Jayawardena & Arunatilake, 2025)

Even more so is the case with home-cooked foods, which tend not to have any details on the nutritional value (Perera, Jayawardena & Arunatilake, 2022). With no proper information, it is challenging to know the nutritional value of what they consume and therefore they find it even harder to eat healthily. This is further compounded by the reality that there are not many apps devoted to non-Western diets. They mostly employ international food databases and are available only in English, failing to capture Sri Lanka's extensive range of ingredients, traditional recipes, and food traditions. Furthermore, these devices simply provide generic nutritional information, without giving custom advice based on age, health status, or individual dietary needs.

Therefore, consumers face significant barriers to realizing the effect their food has on their well-being. Poor available, relevant, and understandable nutritional information impedes their choice towards healthier options, which creates an epidemic of diet-related illnesses across the country (Wijesundara, 2021; Perera, Jayawardena & Arunatilake, 2022).

1.3. Problem Statement

Many Sri Lankans cannot effectively track their diets because they cannot easily interpret complex, multilingual nutrition labels on packaged foods, and current global apps do not support the local context of both packaged and homemade foods, leading to poor nutrition management and increased health risk.

1.4. Research Motivation

Our motivation for this research is to develop a solution that can effectively recognize and interpret Sri Lankan foods, both packaged and traditional, while providing accurate nutritional insights tailored to local diets, as all preexisting models struggle with foods on a local scale. Using datasets from local manufacturers and traditional foods, this system aims to bridge the regional data gap, promote healthier eating habits, and empower users with accessible localized nutritional information.

2. Existing work

Citation	Technology / Algorithm Used	Advantage	Limitation
(Patil, Londhe & Timmapuri, 2025)	CNN-based OCR for nutrition info extraction	Achieved ~75% text extraction accuracy even with low-quality labels; strong barcode recognition	Struggles with complex multilingual or non-standard packaging

(Lodha et al., 2025)	Multi-stage OCR +CNN classification	High ingredient classification accuracy (92.4%); robust for diverse layouts	Limited applicability to local (Sri Lankan) label structures
(Hettiarachchi et al., 2020)	OCR+Text preprocessing on bilingual labels	Highlighted challenges and potential solutions for Sinhala-English packaging	Lacks full automation; dataset limited to experimental samples
(Hen et al., 2023)	Depth prediction CNN fusion model	<10% mean absolute error for calorie prediction	Dataset-specific model; limited generalization to new foods
(Lodha et al., 2025)	OCR+Rule-based normalization	91% ingredient extraction and mapping accuracy	Rule-based system may fail with unseen label formats
(Zhou et al., 2023)	Knowledge graph Embedding model	Improved normalization/linking accuracy by 12%	Requires well-structured ontology; high computational complexity
(University of Moratuwa, 2021)	RNN LSTM Ontology-based diet system	Culturally tailored meal recommendations for Sri Lankan context	Focused on meal planning, not label interpretation

Table 1: Existing Work

3. Contribution to the body of knowledge

3.1. Technological Contribution

This project introduces an AI-based nutritional analysis system designed for Sri Lanka's multilingual and culturally diverse food environment. Unlike Western-focused tools, it combines Computer Vision (CV), Optical Character Recognition (OCR), and Natural Language Processing (NLP) to accurately analyze food images and bilingual (Sinhala, English) labels. It also develops a localized Sri Lankan food database and uses advanced machine learning models to provide personalized diet recommendations. By integrating multi-language OCR, CV, and predictive analytics, the system offers a new AI framework for nutrition tracking in developing and low-resource settings.

3.2. Domain Contribution

This research makes a significant contribution to advancing nutrition awareness and health literacy in Sri Lanka through the application of Artificial Intelligence. It addresses a critical gap by introducing a novel framework to digitize, contextualize, and standardize Sri Lankan food data, an area previously underexplored. The system delivers accurate, accessible, and personalized nutritional insights that empower consumers, support healthcare professionals in clinical diet planning, and inform policymakers on nutrition and food security strategies by integrating advanced AI techniques with local dietary contexts. This project strengthens national efforts to prevent diet-related chronic diseases and promote sustainable, health-conscious eating behaviors.

4. Research Challenges

Multilingual OCR Accuracy – One major challenge is achieving accurate optical character recognition (OCR) for Sinhala and English text. Sri Lankan food labels often include **mixed languages, small fonts and curved or low-contrast text**, which cause errors in text detection and translation during data extraction.

Complex Label Layouts and Visual Distortions – Many food packages contain **irregular layouts, glossy surfaces and overlapping graphics** that interfere with OCR accuracy. Variation in lighting glare and angles during scanning creates distortions that make it harder for the system to isolate nutritional information.

Identification of Plate (Homemade) Food – Recognizing plate-based meals is one of the most difficult challenges. These dishes can **vary in appearance** from one household to another. The system must learn to recognize a large number of Sri Lankan dishes with diverse ingredients, colors and textures which require extensive datasets.

Portion Size and Scale Estimation – Scanning food on a plate, estimating **portion size and food quantity** is difficult because image scaling can vary with distance, angle or lighting without physical measurement tools.

Mixed and Overlapping Food – Sri Lankan meals often contain **multiple items served together on one plate**, such as rice, curries and side dishes. Distinguishing between these mixed or overlapping food poses a challenge for image segmentation and classification, reducing the precision of nutrient estimation.

Limited Local Datasets – There are **no large public datasets** of Sri Lankan packaged or homemade food images for training AI models. Creating such datasets requires extensive manual labeling, standardization, and quality control to ensure balanced coverage of all local food types and label formats.

Health Risk Prediction Reliability – Health risk prediction based on nutrition data depends heavily on **data quality and feature selection**. Inaccurate food recognition or nutrient estimation can lead to **unreliable health predictions**, which may affect user trust and the system’s credibility.

Data Privacy and Ethical Dilemmas – Since the system collects **personal and health-related information**, ensuring privacy, data security, and ethical use of AI is vital. Implementing compliance with privacy regulations and anonymizing sensitive data adds complexity to both development and deployment.

5. Research Aim

The aim of this research is to create an AI powered system that can easily and accurately understand multilingual and complex food labels and scan the nutritional content of home-made food plates providing personalized insights that help people make better food choices tailored to their own health needs and lifestyles, thus supporting healthier habits and enhancing overall well-being.

6. Research Objective

Research Objectives	Explanation	Learning Outcome
Problem Identifying	Identify the key issues Sri Lankan consumers face when interpreting multilingual nutritional labels and tracking their dietary intake.	LO1
Literature Review	RO1. Multilingual Data Extraction & Recognition RO2. Data Normalization and Imputation RO3. Health Risk Prediction Modeling RO4. Contextualized Recommendation Generation RO5. End-to-End System Integration and Feedback	LO1
Data Collecting and Analysing	User Health Inputs: Direct user entries of personal data (age, gender, weight, activity level) and existing medical conditions. Food Image Analysis: Reading multilingual nutrition labels in packaged food and scanning plated food images to detect the nutritional levels using computer vision. Canonical databases (External API): Data fetched from USDA FoodData Central and local Sri Lankan food composition tables used for nutrient standardization.	LO2, LO3
Research Design	Quasi-experimental design is employed to identify the cause-and-effect relationship between users’ nutrient intake and lifestyle metrics (independent variables) and the system’s predicted health risk score (dependent variable).	LO3, LO4

Implementation	Computer Vision and OCR for Food and Nutrition Detection: OCR and Computer Vision to analyze multilingual labels and homemade meals, providing personalized, risk-based nutritional insights. (Home-made meals assumed to a per personal meal scale) Data Extraction, Normalization and Nutrient Estimation: Scalable and accurate nutrition estimation integrates reliable data extraction (OCR/vision) and principled normalization (ontologies and canonical databases like USDA FoodData Central) with predictive models Nutritional Analysis and Interpretation Systems: Nutritional analysis systems convert raw nutrient data from food labels or meal inputs into actionable insights regarding dietary quality, nutrient adequacy, and potential health outcomes. Health Risk Prediction and Personalized Recommendation: Predicting personalized health risks and suitable diet plans from analyzing the user input data and the nutritional level of consumed food on a daily and weekly basis	LO2, LO3, LO4
Testing and Evaluation	To evaluate system performance using accuracy, F1-score, recall, and user feedback, ensuring reliability and usability for local users.	LO2, LO4

Table 2: Research Objective

7. Project Scope

7.1. In-scope

1	Interpret nutritional data on Sri Lankan traditional food
2	Interpret nutritional data on Sri Lankan packaged products
3	Category based identification on broader items
4	OCR and CV as inputs

Table 3: In-Scope

7.2. Out-Scope

1	In-depth datasets on food found outside of Sri Lanka
2	Multilingual Label Translation beyond Sinhala and English
3	Complete analytical accuracy
4	Interpret most regional dishes beyond Sri Lanka

Table 4: Out-Scope

7.3. Feature Prototype

Process 01 – Computer Vision and OCR for Food and Nutrition Detection

The process begins with users capturing or uploading an image of either a packaged product or a meal plate using their smartphone camera or gallery. The system supports common image formats like JPG and PNG and automatically enhances the image by adjusting brightness, contrast, resizing, and reducing noise to ensure clear quality analysis. Upon uploading, the system automatically determines whether the image is of a packaged food label or a meal plate by analyzing visual features such as text regions, barcodes, or natural food textures, and routes the image to the appropriate analysis pipeline. For packaged products, the system identifies and isolates the Nutrition Facts section by cropping out branding and unrelated background elements, focusing solely on the relevant label. Optical Character Recognition (OCR) is then applied to the cropped label to extract printed nutritional information such as energy, sugar, fat, carbohydrates, and protein, with image pre-processing steps like grayscale conversion, thresholding, and denoising used to improve accuracy for multilingual labels in Sinhala and English. The raw extracted text is then forwarded to the next step for data extraction, normalization, and nutrient estimation. In contrast, if the image is recognized as a meal plate, the system classifies the dish type using a classifier trained on locally collected Sri Lankan meal photos, such as rice and curry, dhal curry, and fish curry. The identified food name and confidence score are then passed to the nutrient estimation stage, completing the nutritional analysis process.

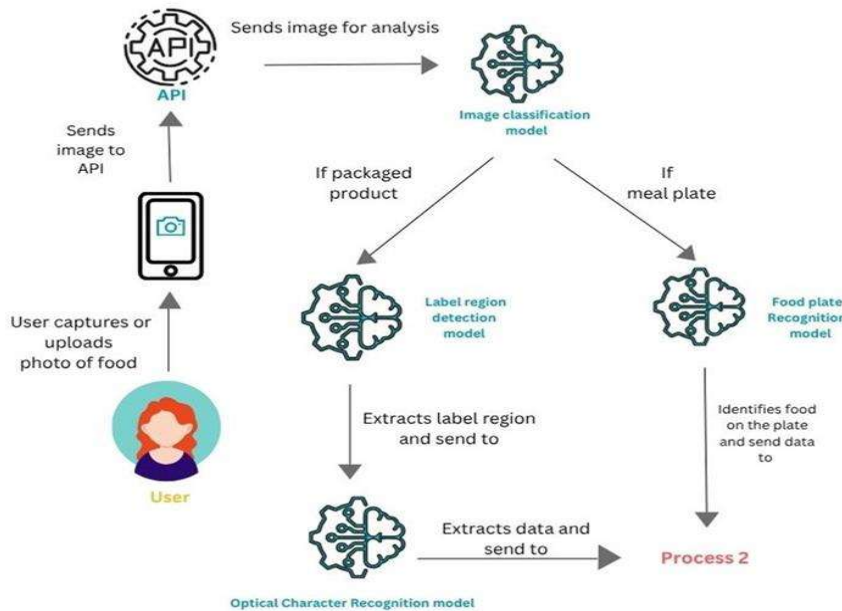


Figure 1: Feature Prototype Diagram for Component 1

Process 02 - Data Extraction, Normalization and Nutrient Estimation

After image-based food and text recognition, the data obtained is processed through text parsing, normalization, and nutrient estimation to give accurate, comparable nutrient information. The OCR text from food labels with their wavy text is initially processed through Named Entity Recognition (NER) and rule-based parsing to obtain high-level nutrients like protein, fat, and carbohydrates along with their values and units, correcting spelling mistakes in food names. Next, the extracted entities are mapped into typed formats such as JSON or tables with mandated fields such as name of nutrient, value, unit, serving size, and type of source. Validation rules check for consistency errors, automatically fixing or flagging incorrect data. Next, ontology-based normalization maps the food names and ingredients onto standard food ontologies and databases, normalizing labeling across sources, removing regional naming conventions, and scaling units to per 100 g or per serving. Statistical imputation and machine learning regression models impute the missing micronutrient or serving data when it is not present by combining features from labels, food categories, and visual aspects such as portion size and texture. Predictive models constructed from harmonized nutrient databases provide precise estimates for diet evaluation. Finally, the process creates a fully normalized nutrient profile for each food item, including macro- and micronutrient values, serving details, confidence scores, and data sources, ready for further analysis in the next process.

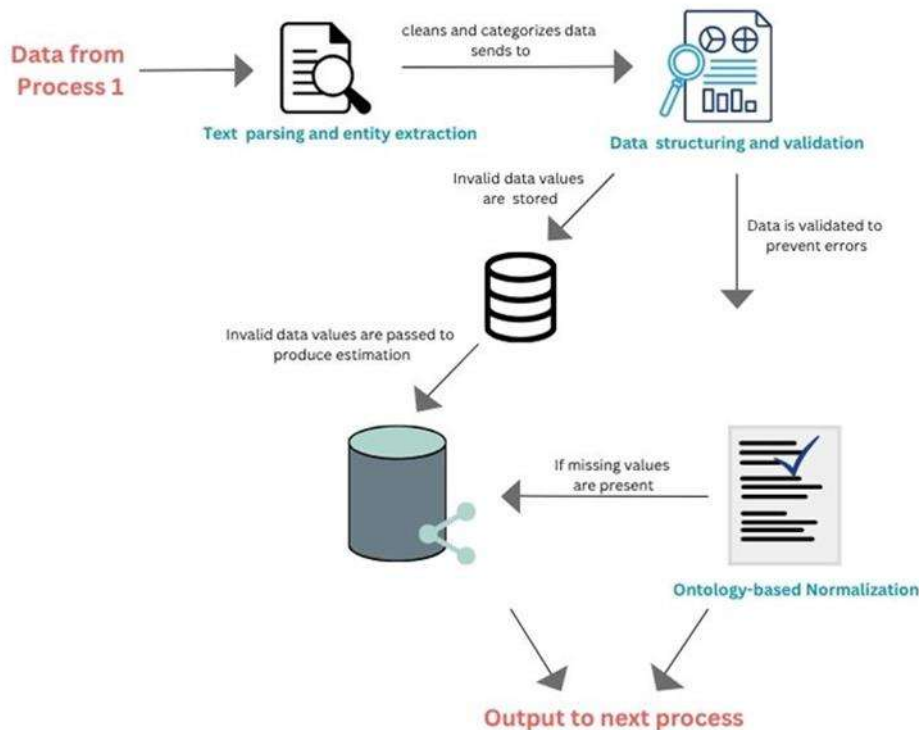


Figure 2: Feature Prototype Diagram for Component 2

Process 03 - Nutritional Analysis and Interpretation Systems

Following nutrient estimation in Process 02, this stage transforms standardized nutrient data into meaningful dietary insights by evaluating both individual meals and cumulative intake patterns. The system first performs nutrient profiling and caloric breakdown, calculating total energy and macronutrient distribution and benchmarking them against FAO/WHO and Sri Lankan RDAs to identify energy-dense, protein-rich, or fat-heavy foods. It then conducts micronutrient adequacy evaluation, assessing key nutrients such as Iron, Calcium, Vitamin C, and Sodium relative to daily requirements and computing % Daily Values(%DV) per serving to flag excesses or deficiencies. Next, the system applies diet quality and health index scoring through composite indices such as Nutrient Density Score (NDS), Healthy Eating Index (HEI)-like score, and Sodium-Sugar-Saturated Fat (SSSF) Score, each normalized on a 0-100 scale for intuitive interpretation. In the comparative and cumulative analysis stage, nutrient data from multiple meals are aggregated to generate daily or weekly summaries, allowing users to compare their consumption with dietary goals using visual analytics for improved self-awareness. Finally, the system performs data integration for health risk modelling, producing a structured dataset containing nutrient ratios, quality scores, and adequacy metrics that feed into Process 04 to enhance health risk prediction and personalized recommendations.

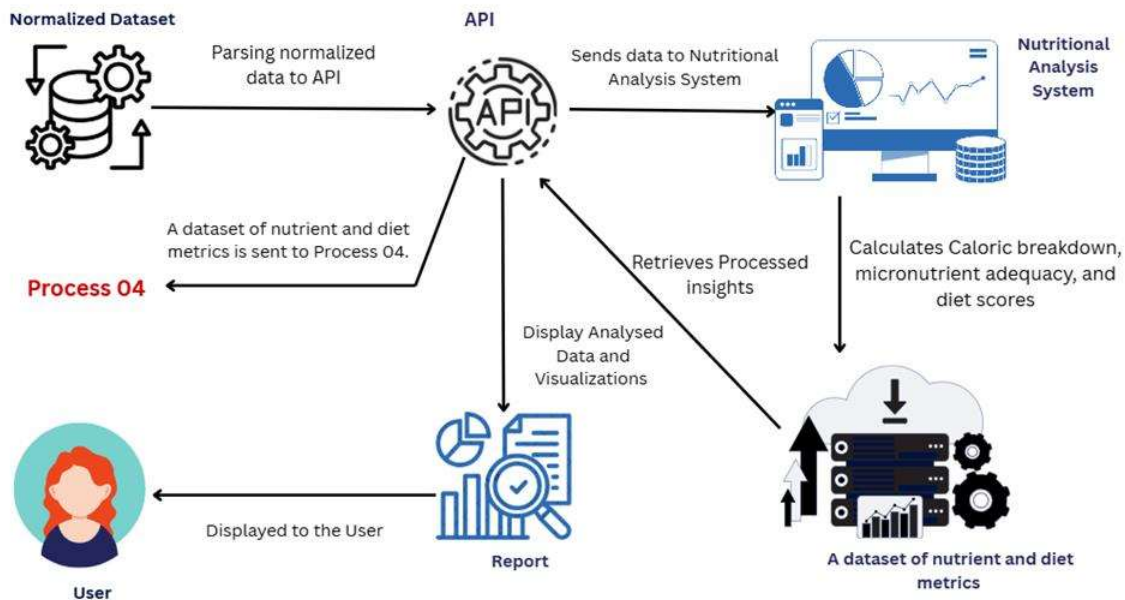


Figure 3: Feature Prototype Diagram for Component 3

Process 04 – Health Risk Prediction and Personalized Recommendation

Patients input personal and health data such as age, gender, height, weight, activity level, and conditions like diabetes or hypertension, and food label nutrition data entered from food product barcodes scanned on their phone or meal photographs, all stored securely for processing. The program cleans and normalizes the data, adjusts nutrient levels with USDA and local database sources, and fills in missing values with machine learning. It combines nutrition intake, demographics, and lifestyle information to predict health risks like Type 2 Diabetes, Hypertension, Obesity, and cardiovascular disease, and gives each user a numeric risk factor within Low, Moderate, or High. Informed by these predictions, personalized dietary recommendations are generated with large language models and ontology-based inferencing, tailored to Sri Lankan eating habits. For example, the system may advise high-risk hypertension users to reduce sodium intake using low-sodium alternatives. All users are able to view their risk scores, nutrient breakdowns, and week-to-week progress on a dashboard with basic graphics and actionable feedback. The system refreshes health profiles regularly as new information about foods is incorporated, automatically modifying recommendations to reflect changing nutrition and risk trends.

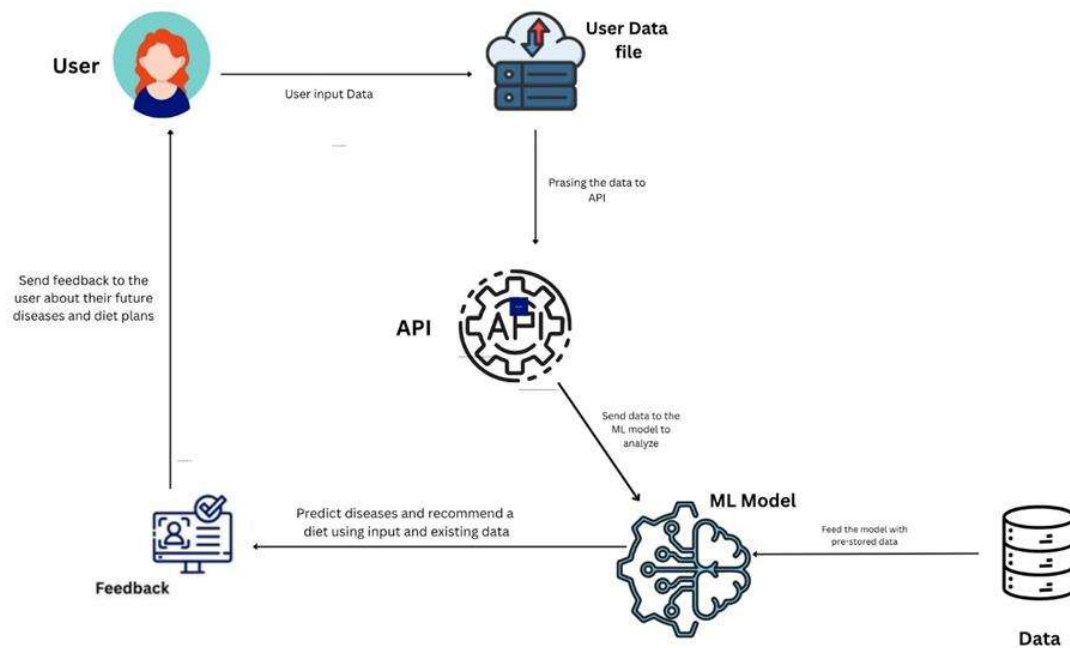


Figure 4: Feature Prototype Diagram for Component 4

Nutri Scanner System Diagram

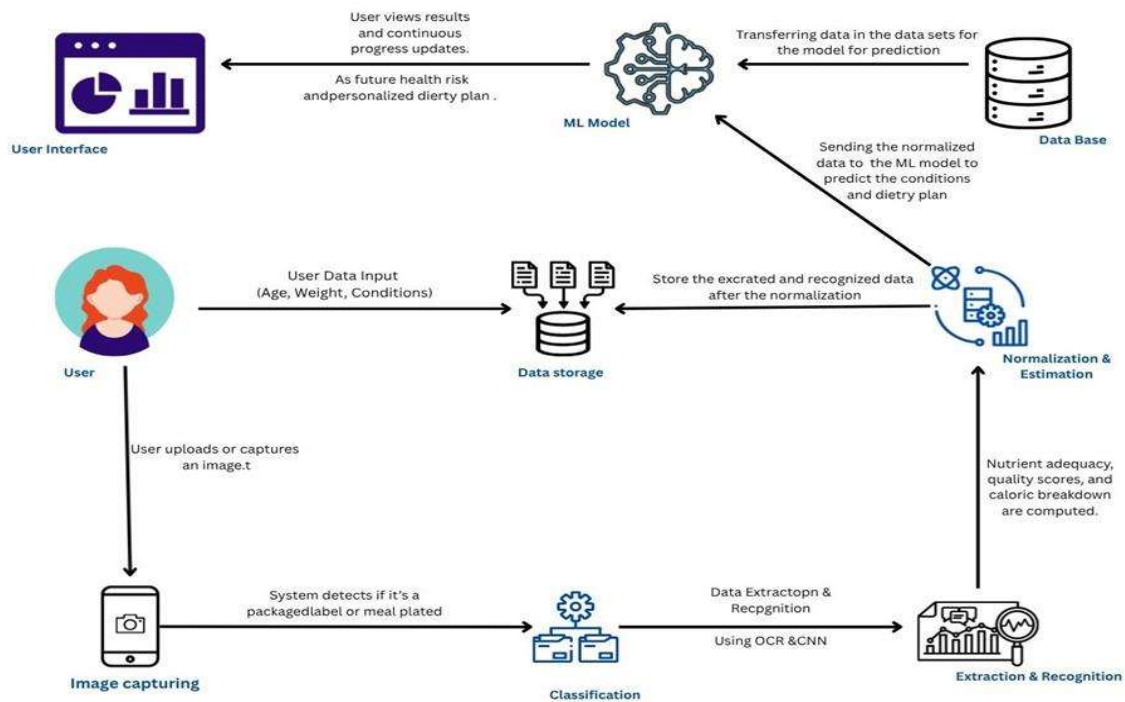


Figure 5: Full System Diagram

8. Methodology

8.1. Research Methodology

Research Philosophy	We will use a positive methodology since our project relies on quantitative data and measurable outcomes such as nutrient extraction accuracy, computational performance, and system usability metrics.
Research Approach	A deductive approach will be adopted, drawing on established nutrition guidelines, machine learning principles, and prior research on nutrition apps to guide system development and shape hypotheses.
Research Strategy	The main approach will be experimental and computational, using datasets to train, test, and validate the system. Performance evaluation metrics such as

	accuracy, speed, and clarity of outputs will be collected and statistically analyzed to assess system effectiveness.
Research Choice	A single research approach will be adopted, using a quantitative strategy to ensure that data collection, storage, and analysis are guided by measurable outcomes and statistical evidence rather than subjective interpretation.
Time Zone	A combined cross-sectional and longitudinal approach will be used. The cross-sectional component evaluates the system's functionality and performance after prototype development, while the longitudinal aspect captures and analyzes user data over time through daily and weekly reports to observe nutritional trends.

Table 5: Research Methodology

8.2. Development Methodology

8.2.1. Life Cycle Model: The Agile Software Development Life Cycle Model will be used for this project because the proposed system, Nutri Scanner an AI-powered nutrition tracking and health risk prediction application, includes modules such as OCR, nutrient analysis, health risk prediction, and recommendation generation. Agile enables the team to develop and test each module in short, iterative sprints, incorporate continuous feedback from supervisors and users, quickly adapt to new requirements or changes, and maintain parallel progress across multiple components like the front-end, database, and AI models.

8.2.2. Design Methodology: The Object-Oriented Analysis and Design (OOAD) approach will be used because the system includes multiple entities such as User, Food Item, Nutrition Label, Health Risk Model, and Recommendation Engine that naturally fit into an object-oriented structure. OOAD helps simplify system complexity by modeling real-world entities as objects, supports modularity and reusability for future expansion, and enables a clear mapping between design and implementation.

8.2.3. Evaluation Methodology: The evaluation process will be performance, correctness, and user satisfaction of each important component. The project is evaluated against five basic components. The OCR Module is approximated using Text Recognition Accuracy to provide accurate information extraction from nutrition labels. The Nutrient Estimation accuracy is quantified by using the Mean Absolute Error (MAE) and RMSE, providing a measure of how close estimated nutrient values are to real values. The Health Risk Prediction model performance is rigorously validated using typical machine learning metrics like Accuracy, Precision, Recall, and F1-Score. Finally, the system's usability is measured through the Recommendation System, as Top N recommendation accuracy and user opinion surveys, and overall System Usability, and ascertained by user responses and qualitative testing for ease and local utility.

8.3. Project Management Methodology

8.3.1. Gantt Chart

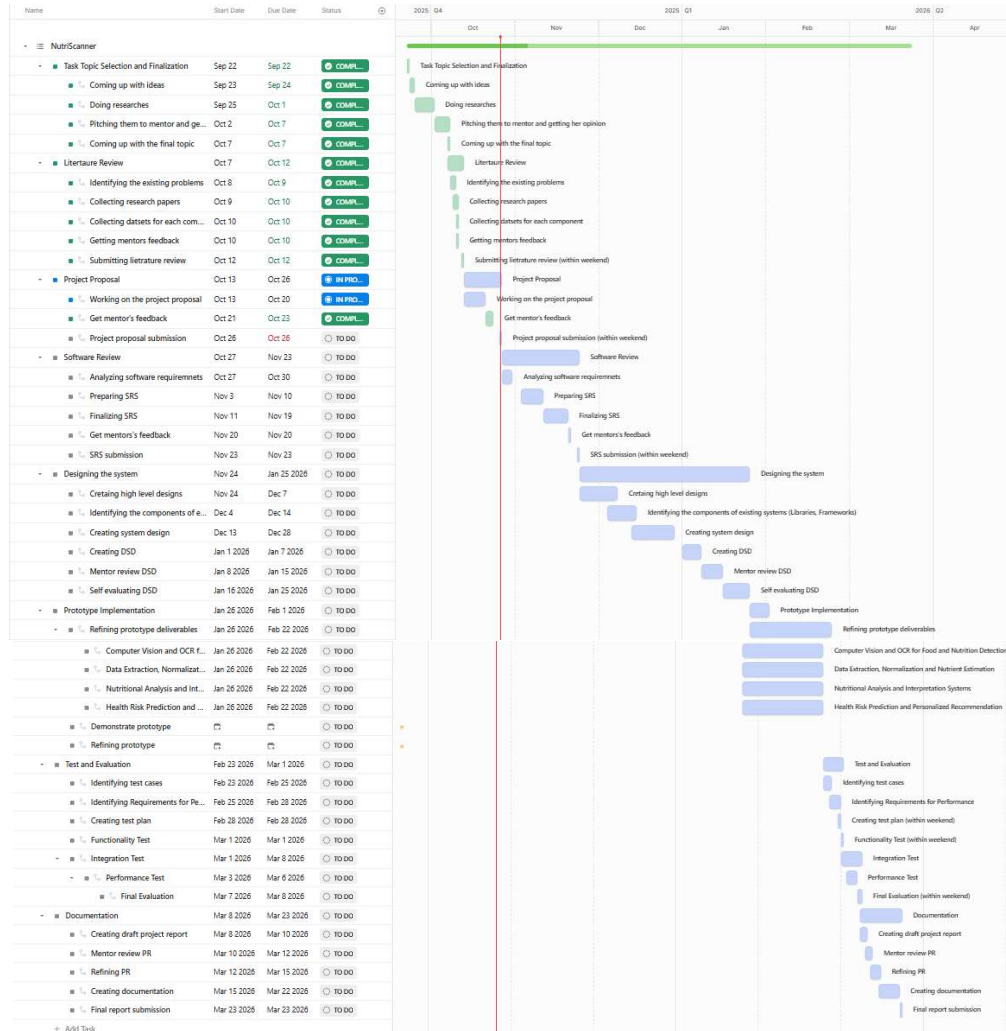


Figure 6: Gantt Chart

8.3.2. Deliverables

Deliverable	Date
Semester 01	
Literature review submit	Week 03
Project Proposal submit to the supervisor	Week 04

Final Project Proposal submit	Week 05
SRS submit to the supervisor	Week 08
Final SRS submit	Week 09
Semester 02	
Implementation of Prototype	Week 14
Testing and Evaluation	Week 19
Submission of documentation and final report	Week 23

Table 6: Deliverables

8.3.3. Resource Requirements

Hardware Requirements

A high-performance CPU like the Intel Core i7 9th gen or Ryzen 7 (or higher) is required to handle intensive computational tasks and ensure smooth system performance. The system should have at least 16GB of RAM to efficiently process large datasets and run machine learning algorithms. For storage, a minimum of 250GB SSD or 1TB HDD is needed to accommodate datasets, images and model files.

Software Requirements

Python is used for system development, offering robust error management and extensive libraries. PyCharm helps create and manage project applications, while Jupyter Notebook and Google Colab support development, testing, and model training. Databases are managed using SQLite, MySQL, or PostgreSQL. The front-end interface is built with HTML, CSS, and JavaScript and backend development with PHP or Node.js handles database interactions. Machine learning models are developed and trained using TensorFlow and PyTorch. Windows or Mac OS platforms support high-performance computing and data processing. MS Word is used for creating reports and documentation, and SPSS or R are employed for statistical analysis.

Data Requirements

The project will use both global datasets and local Sri Lankan data. Global datasets help create an accurate and generalized model applicable in any setting, while local food data ensures relevance for Sri Lankan dietary habits. This includes nutritional tables, multilingual food labels, food images, and simulated user profiles.

Skill Requirements

They key skills required for this project include data collection and dataset searching to gather relevant and sufficient information for machine learning, programming and machine learning expertise to develop models and process data, and statistical analysis and data visualization to interpret and present findings effectively. Problem-solving and critical thinking are essential to address challenges and optimize solutions throughout the project. Additionally, effective time management ensures that tasks and milestones are met efficiently, while writing reports and documentation is crucial for clear communication and record-keeping. Planning and scheduling help organize the workflow and coordinate activities for successful project execution. These skills collectively support the development and implementation of a robust AI -powered nutrition tracking system.

8.3.4. Risk Management

Risk Item	Severity	Frequency	Mitigation Plan
Unavailable or Incomplete Data	5	4	Use open global nutrition datasets and collaborate with local sources for Sri Lankan food data. Apply data normalization techniques to fill missing values.
Low Accuracy in OCR/NLP Models	4	4	Fine-tune AI models on local multilingual data (Sinhala, English). Continuously test and retrain using feedback datasets.
User Misinterpretation of Results	3	3	Provide clear, culturally relevant explanations and use color-coded indicators to make nutritional risk easy to understand.
Hardware/Software Failure	5	3	Store data and models in cloud platforms (Google Colab, GitHub, Google Drive). Maintain regular backups and use version control for code.

Table 7: Risk Management

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